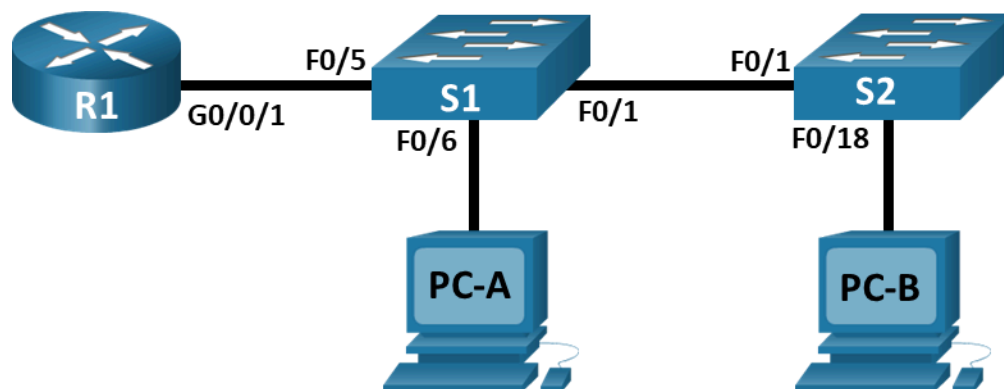


NAME: CARBONEL, NICOLE D.G.	DATE: MARCH 19, 2026
SECTION: BSCPE 3-4	PROF: DR. LUTZER U. REYES

Packet Tracer - Troubleshoot Inter-VLAN Routing – Physical Mode

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/1.3	10.3.0.1	255.255.255.0	N/A
	G0/0/1.4	10.4.0.1	255.255.255.0	
	G0/0/1.13	10.13.0.1	255.255.255.0	
S1	VLAN 3	10.3.0.11	255.255.255.0	10.3.0.1
S2	VLAN 3	10.3.0.12	255.255.255.0	10.3.0.1
PC-A	NIC	10.4.0.50	255.255.255.0	10.4.0.1
PC-B	NIC	10.13.0.50	255.255.255.0	10.13.0.1

VLAN Table

VLAN	Name	Interface Assigned
3	Management	S1: VLAN 3 S2: VLAN 3

Packet Tracer - Troubleshoot Inter-VLAN Routing – Physical Mode

4	Operations	S1: F0/6
7	ParkingLot	S1: F0/2-4, F0/7-24, G0/1-2 S2: F0/2-17, F0/19-24, G0/1-2
8	Native	N/A
13	Maintenance	S2: F0/18

```
S1>ena
Password:
S1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
3	Management	active	
4	Operations	active	Fa0/6
7	ParkingLot	active	Fa0/2, Fa0/3, Fa0/4, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
8	Native	active	
13	Maintenance	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
S1#
```

```
S2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
3	Management	active	
4	Operations	active	
7	ParkingLot	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
8	Native	active	
13	Maintenance	active	Fa0/18
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
S2#
```

Objectives

Part 1: Evaluate Network Operation

Part 2: Gather Information, Create an Action Plan, and Implement Corrections

Background / Scenario

All the network devices in this Packet Tracer Physical Mode (PTPM) activity have been preconfigured to include intentional errors that are keeping the inter-VLAN routing from working. Your task is to evaluate the network, identify, and correct the configuration errors to restore full inter-VLAN connectivity. You may find errors with the configurations which are not directly related to inter-VLAN routing. These errors impact the ability of the network devices to perform this function.

Note: The design approach used in this activity is to assess your ability to configure and troubleshoot inter-VLAN routing only. This design may not reflect networking best practices.

Instructions

Part 1: Evaluate Network Operation

Requirements:

No VLAN 7 traffic is allowed on the trunks because there are no devices in VLAN 7.

VLAN 8 is the native VLAN.

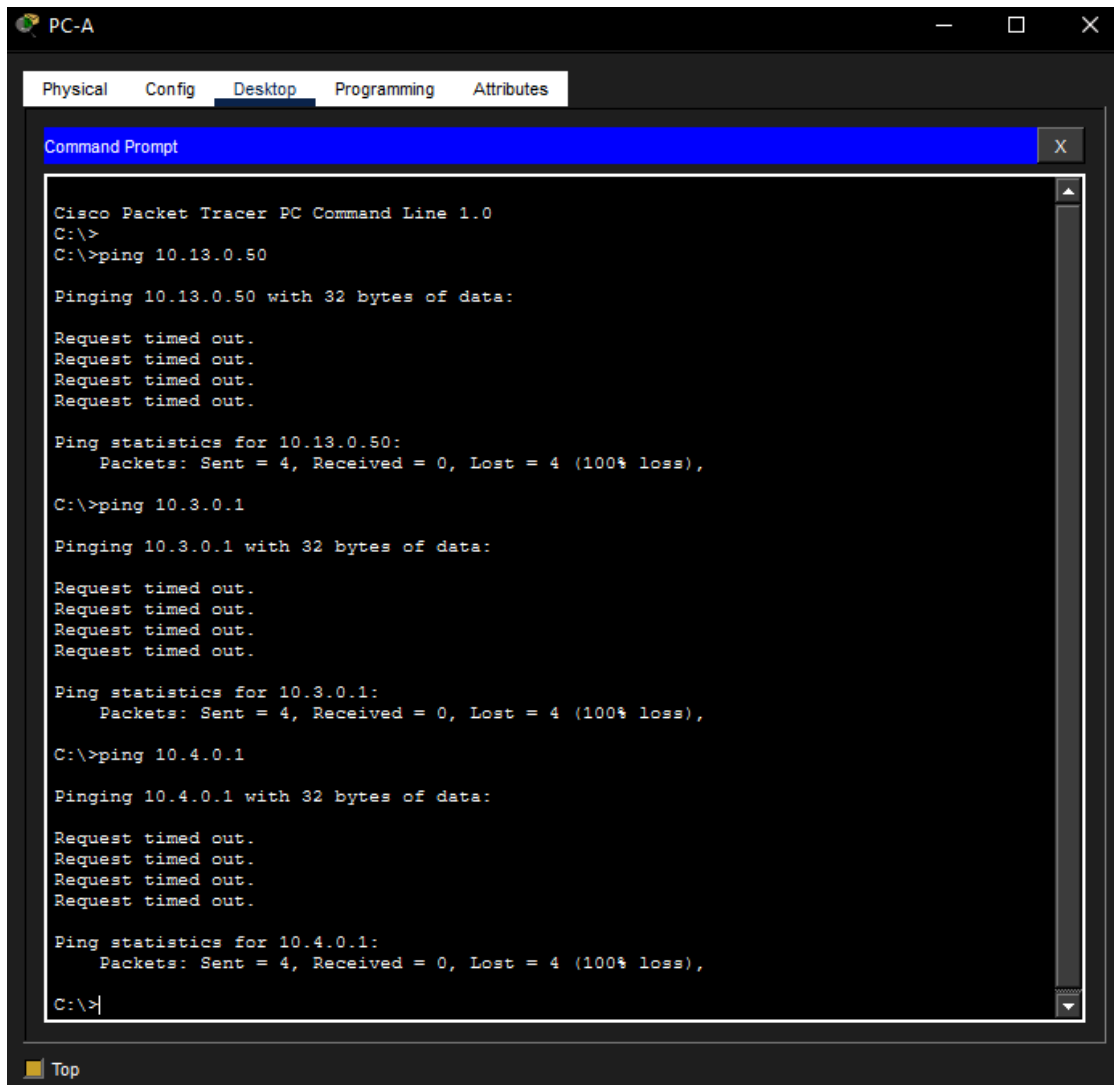
All trunks are static.

End to end connectivity.

- Use the laptop computer and appropriate cable to console into the network devices for testing and configuration purposes. The login password on all network devices is “**cisco**” and the enable password is “**class**”. You can click and drag the console connection from the console port of one device to another, but you will have to start a new terminal session.
- Use the **ping** command to test the following criteria and record the results in the table below.

From	To	Ping Results
R1	S1 VLAN 3 (10.3.0.11)	Unsuccessful
	S2 VLAN 3 (10.3.0.12)	Unsuccessful
	PC-A (10.4.0.50)	Unsuccessful
	PC-B (10.13.0.50)	Unsuccessful
S1	S2 VLAN 3 (10.3.0.12)	Unsuccessful
	PC-A (10.4.0.50)	Unsuccessful
	PC-B (10.13.0.50)	Unsuccessful
S2	PC-A (10.4.0.50)	Unsuccessful
	PC-B (10.13.0.50)	Unsuccessful

PC-A



The screenshot shows the PC-A interface with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.13.0.50

Pinging 10.13.0.50 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.13.0.50:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.3.0.1

Pinging 10.3.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.3.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.4.0.1

Pinging 10.4.0.1 with 32 bytes of data:

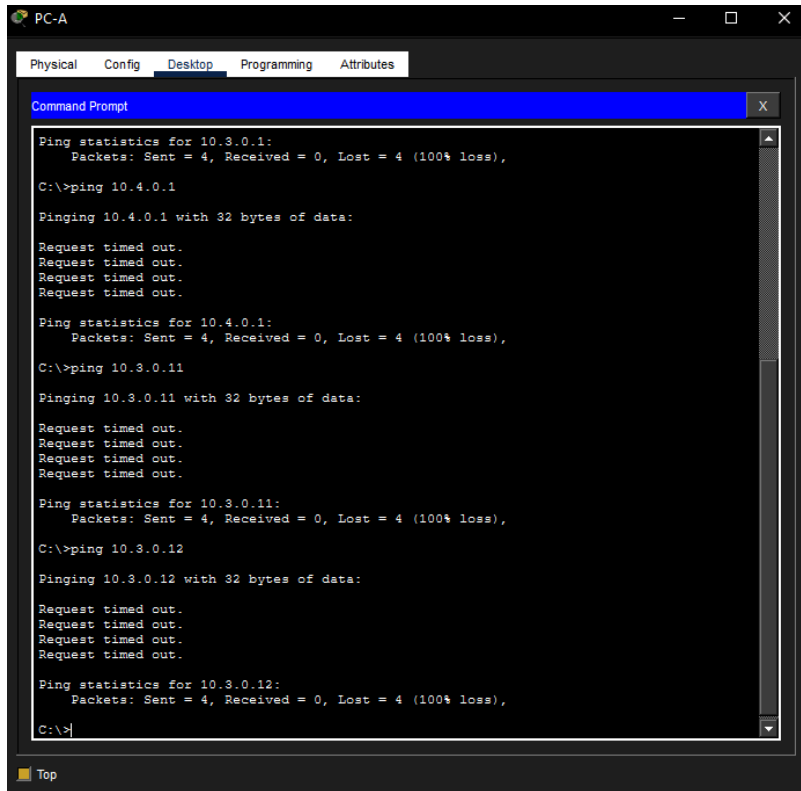
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.4.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

At the bottom left of the window, there is a 'Top' button.

Packet Tracer - Troubleshoot Inter-VLAN Routing – Physical Mode



```
PC-A
Physical Config Desktop Programming Attributes
Command Prompt
Ping statistics for 10.3.0.1:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.4.0.1

Pinging 10.4.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.4.0.1:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.3.0.11

Pinging 10.3.0.11 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.3.0.11:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.3.0.12

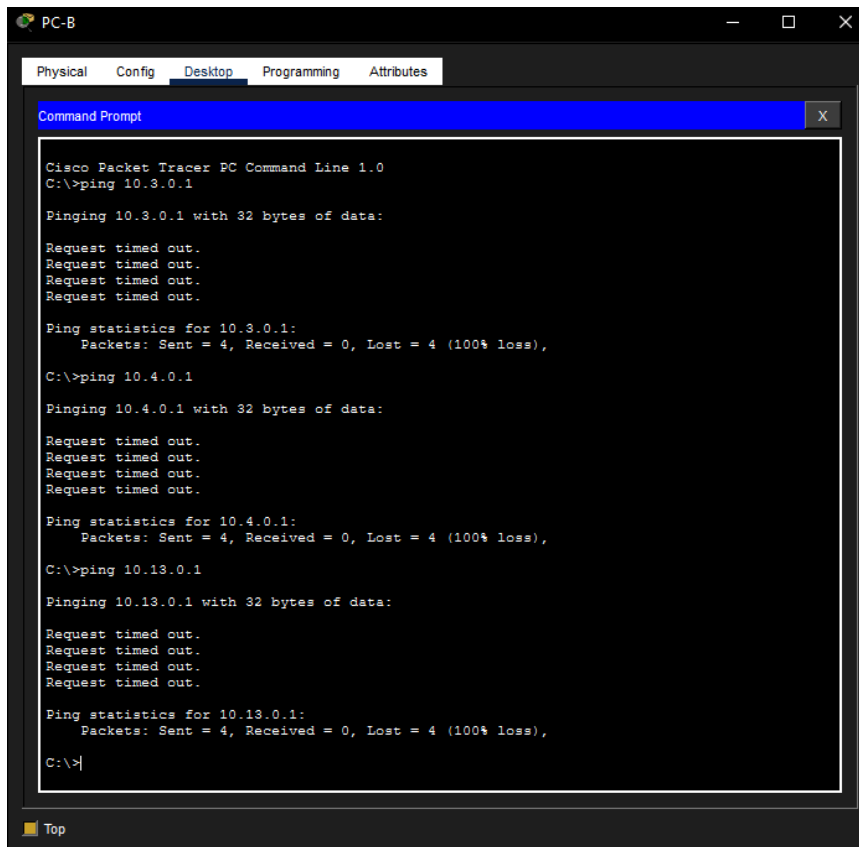
Pinging 10.3.0.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.3.0.12:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC-B



```
PC-B
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.3.0.1

Pinging 10.3.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.3.0.1:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.4.0.1

Pinging 10.4.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.4.0.1:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

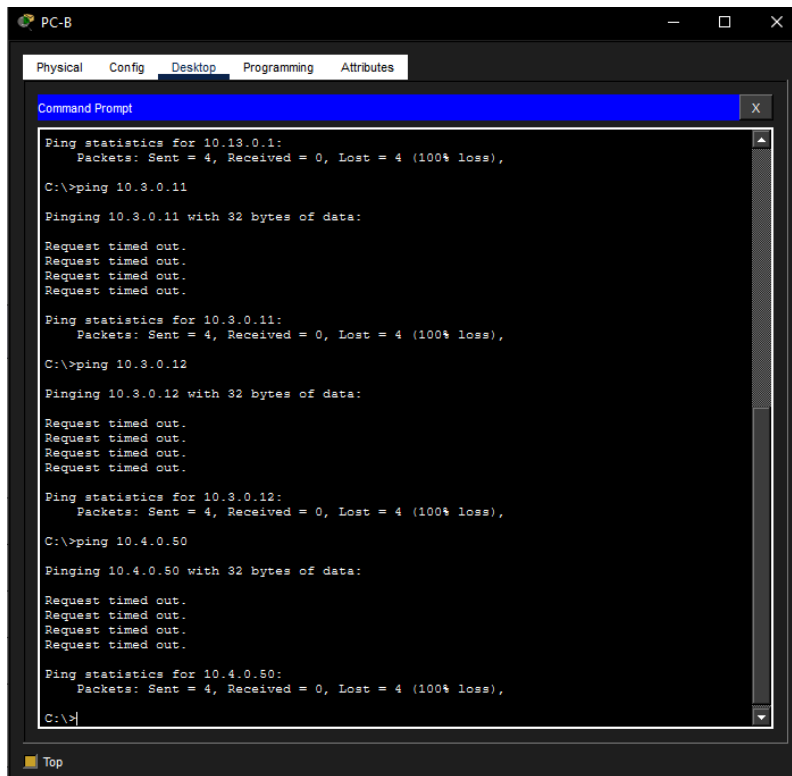
C:\>ping 10.13.0.1

Pinging 10.13.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.13.0.1:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```



The screenshot shows the Command Prompt window for PC-B. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt displays the following text:

```
Command Prompt
Ping statistics for 10.13.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.3.0.11

Pinging 10.3.0.11 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.3.0.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.3.0.12

Pinging 10.3.0.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.3.0.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.4.0.50

Pinging 10.4.0.50 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.4.0.50:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```



The screenshot shows the CLI window for Router1. The window has tabs for Physical, Config, CLI, and Attributes, with CLI selected. The CLI displays the following text:

```
Router1
Press RETURN to get started!

Press RETURN to get started!

Press RETURN to get started!

Press RETURN to get started!

Authorized Access Only!
User Access Verification
Password:
R1>ena
Password:
R1#ping 10.3.0.11

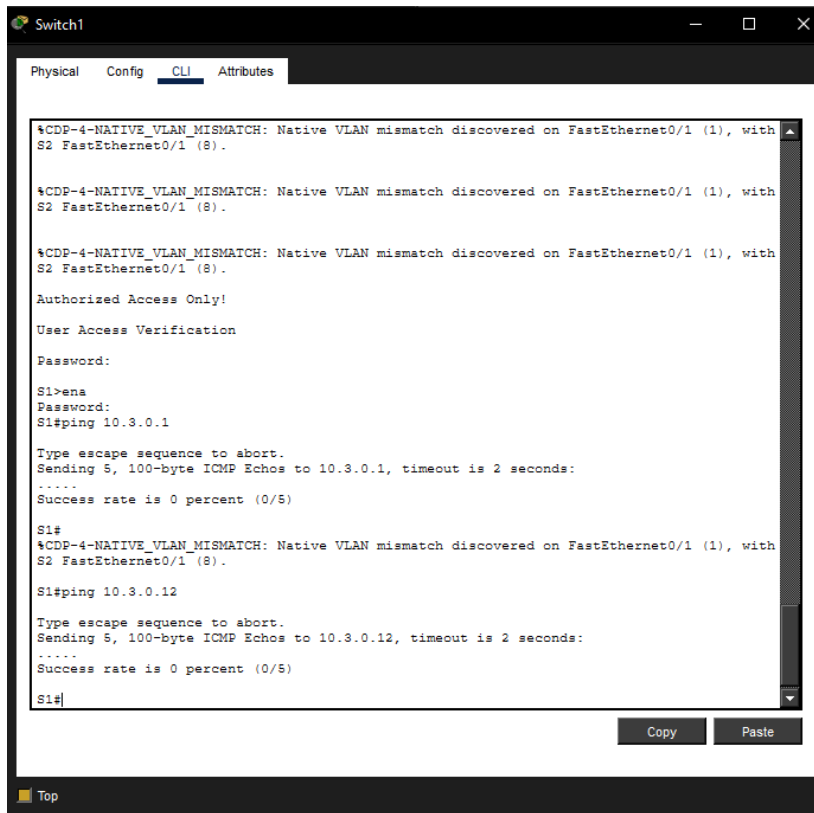
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.0.11, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R1#ping 10.3.0.12

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.0.12, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R1#
```

Switch 1 and 2



Switch1

Physical Config CLI Attributes

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with
S2 FastEthernet0/1 (8).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with
S2 FastEthernet0/1 (8).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with
S2 FastEthernet0/1 (8).

Authorized Access Only!

User Access Verification

Password:
S1>ena
Password:
S1#ping 10.3.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.0.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

S1#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with
S2 FastEthernet0/1 (8).

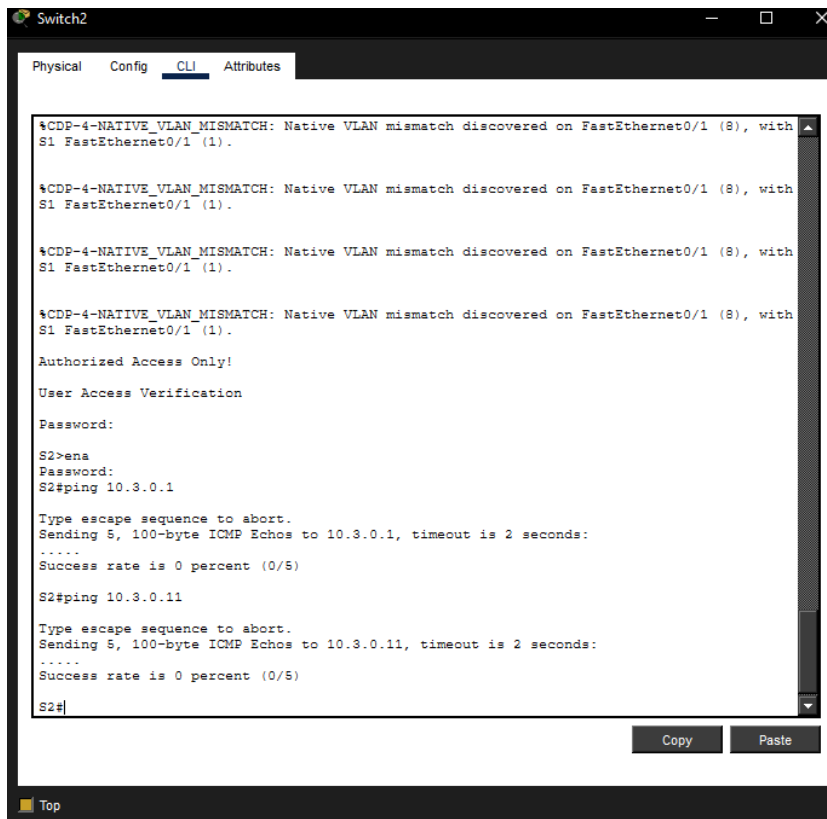
S1#ping 10.3.0.12

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.0.12, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

S1#
```

Copy Paste

Top



Switch2

Physical Config CLI Attributes

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (8), with
S1 FastEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (8), with
S1 FastEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (8), with
S1 FastEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (8), with
S1 FastEthernet0/1 (1).

Authorized Access Only!

User Access Verification

Password:
S2>ena
Password:
S2#ping 10.3.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.0.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

S2#ping 10.3.0.11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.0.11, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

S2#
```

Copy Paste

Top

Packet Tracer - Troubleshoot Inter-VLAN Routing – Physical Mode

Network diagram showing a topology with Router1 (2911) connected to Switch1 (2960) via GigabitEthernet0/1 and Fa0/5. Switch1 is connected to Switch2 (2960) via Fa0/1 and Fa0/18. Switch1 has PC-A connected to Fa0/6, and Switch2 has PC-B connected to Fa0/18. The interface Fa0/6 on Switch1 is highlighted in red, indicating a problem.

Below the diagram is a toolbar with icons for various devices and a 'Scenario 0' dropdown menu. To the right is a table showing the status of various components:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Failed	Router1	Switch1	ICMP		0.000	N	0	(e...		(delete)
Failed	Router1	Switch2	ICMP		0.000	N	1	(e...		(delete)
Failed	Router1	PC-A	ICMP		0.000	N	2	(e...		(delete)
Failed	Router1	PC-B	ICMP		0.000	N	3	(e...		(delete)
Failed	Switch1	Switch2	ICMP		0.000	N	4	(e...		(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Failed	Router1	Switch1	ICMP		0.000	N	0	(e...		(delete)
Failed	Router1	Switch2	ICMP		0.000	N	1	(e...		(delete)
Failed	Router1	PC-A	ICMP		0.000	N	2	(e...		(delete)
Failed	Router1	PC-B	ICMP		0.000	N	3	(e...		(delete)
Failed	Switch1	Switch2	ICMP		0.000	N	4	(e...		(delete)

Network diagram showing the same topology as above, but with a different configuration. The interface Fa0/6 on Switch1 is now highlighted in green, indicating it is operational. The interface Fa0/18 on Switch2 is highlighted in red, indicating a problem.

Below the diagram is a toolbar with icons for various devices and a 'Scenario 0' dropdown menu. To the right is a table showing the status of various components:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Failed	Switch1	Switch2	ICMP		0.000	N	4	(e...		(delete)
Failed	Switch1	PC-A	ICMP		0.000	N	5	(e...		(delete)
Failed	Switch1	PC-B	ICMP		0.000	N	6	(e...		(delete)
Failed	Switch2	PC-A	ICMP		0.000	N	7	(e...		(delete)
Failed	Switch2	PC-B	ICMP		0.000	N	8	(e...		(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Failed	Switch1	Switch2	ICMP		0.000	N	4	(e...		(delete)
Failed	Switch1	PC-A	ICMP		0.000	N	5	(e...		(delete)
Failed	Switch1	PC-B	ICMP		0.000	N	6	(e...		(delete)
Failed	Switch2	PC-A	ICMP		0.000	N	7	(e...		(delete)
Failed	Switch2	PC-B	ICMP		0.000	N	8	(e...		(delete)

Part 2: Gather Information, Create an Action Plan, and Implement Corrections

- a. For each requirement that is not met, gather information by examining the running configuration and the routing tables to develop a hypothesis for what is causing the malfunction.
 - Trunking Misconfiguration: The connection between S1 and S2 was not set to trunk mode, preventing VLAN traffic (Management and Maintenance) from passing through.
 - Native VLAN Mismatch: S1 was set to Native VLAN 8, but S2 was likely still on the default (VLAN 1).
 - Missing VLANs: VLAN 13 was not defined in the switch database on S2, so the port couldn't carry that traffic.
 - PC Firewalls: The Windows Firewalls on PC-A and PC-B were blocking ICMP echo requests (pings).
- b. Create an action plan that you think will fix the issue. Develop a list of all the commands you intend to issue to fix the issue, and a list of all the commands you need to revert the configuration, should your action plan fail to correct the issue.

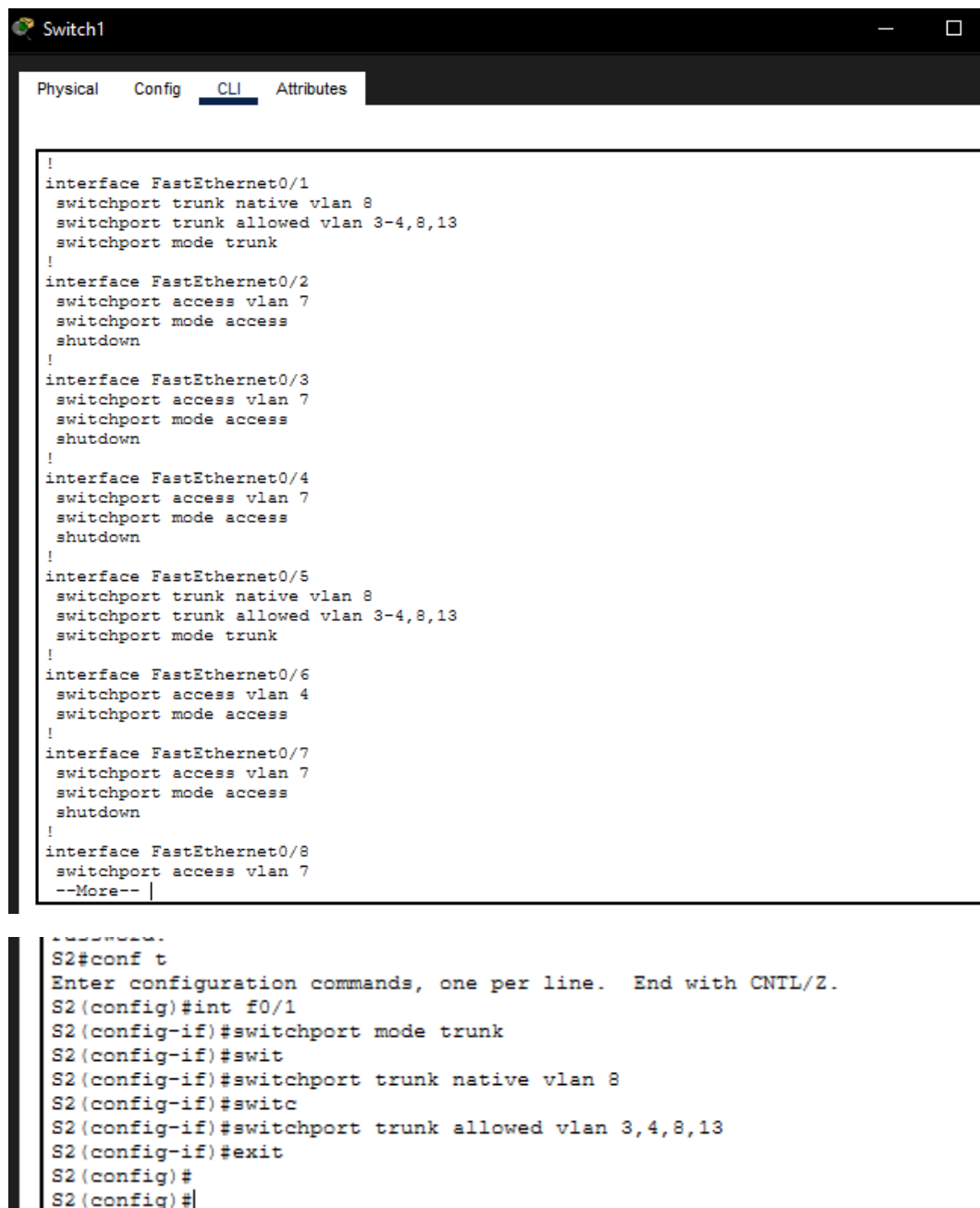
Hint: If you need to reset a switchport to default configuration, use the **default interface** *interface name* command.

As an example for F0/10:

S1(config)# **default interface f0/10**

```
R1(config)#interface g0/1.8
R1(config-subif)#encapsulation dot1q 8 native
R1(config-subif)#exit
R1(config)#
```

```
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#int f0/1
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk native vlan 8
S1(config-if)#swit
S1(config-if)#switchport trunk allowed vlan 3,4,8,13
S1(config-if)#exit
S1(config)#vlan 13
S1(config-vlan)#name Maintenance
S1(config-vlan)#exit
S1(config)#
```



```

Switch1
Physical Config CLI Attributes

!
interface FastEthernet0/1
 switchport trunk native vlan 8
 switchport trunk allowed vlan 3-4,8,13
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 7
 switchport mode access
 shutdown
!
interface FastEthernet0/3
 switchport access vlan 7
 switchport mode access
 shutdown
!
interface FastEthernet0/4
 switchport access vlan 7
 switchport mode access
 shutdown
!
interface FastEthernet0/5
 switchport trunk native vlan 8
 switchport trunk allowed vlan 3-4,8,13
 switchport mode trunk
!
interface FastEthernet0/6
 switchport access vlan 4
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 7
 switchport mode access
 shutdown
!
interface FastEthernet0/8
 switchport access vlan 7
--More--

S2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#int f0/1
S2(config-if)#switchport mode trunk
S2(config-if)#swit
S2(config-if)#switchport trunk native vlan 8
S2(config-if)#switc
S2(config-if)#switchport trunk allowed vlan 3,4,8,13
S2(config-if)#exit
S2(config)#
S2(config)#

```

- c. Execute your action plans one at a time for each criterion that fails and record the fix actions.

To resolve the connectivity issues, I first converted all inter-switch and router-switch links to static trunk mode using the switchport mode trunk command to ensure consistent frame tagging across the topology. I then aligned the native VLAN by configuring VLAN 8 as the native VLAN on S1, S2, and the R1 sub-interface, which prevented traffic leaks and resolved Spanning Tree Protocol errors. To comply with the security requirements, I implemented VLAN pruning using the switchport trunk allowed vlan command to explicitly exclude VLAN 7 traffic from the trunks. I restored Inter-VLAN routing by configuring R1 with the correct sub-interfaces and 802.1Q encapsulation, acting as the functional gateway for the Operations and Maintenance VLANs. Finally, I secured the network by assigning all unused ports to VLAN 7 and administratively disabling them with the shutdown command, ultimately verifying that PC-A could successfully ping PC-B and all infrastructure devices could reach their default gateways.

Router

```
---
R1#ping 10.3.0.11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.0.11, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#ping 10.3.0.12

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.0.12, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#ping 10.4.0.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.4.0.50, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#ping 10.13.0.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.13.0.50, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#
```

Switch

```
S1#ping 10.3.0.12

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.0.12, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

S1#ping 10.4.0.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.4.0.50, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

S1#ping 10.13.0.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.13.0.50, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

S1#
```

```
S2#ping 10.4.0.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.4.0.50, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

S2#ping 10.13.0.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.13.0.50, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

S2#
```